

EduAnimAI Project Overview

Project summary: EduAnimAI is an AI-powered educational animation generator that converts text into multimedia. It supports math, science, history, and other topics by generating diagrams, characters, maps, objects, and more.
Repository: [Snavol/EduAnimAI](https://github.com/Snavol/EduAnimAI)

1. Tech stack

Language: TypeScript, HTML

Frontend: React 19 with Vite

AI: Google Gemini via `@google/genai`, using `gemini-2.5-flash`

Animation: Framer Motion for movement, transitions, zoom, and scene playback

Icons/UI: lucide-react and inline JSX utility styling

2. How the app works

`App.tsx` is the main UI. It stores the prompt, triggers Gemini generation, and renders the response with an `AnimationScript`. When a user submits a topic, `handleGenerate` calls `generateAnimationScript` from `services/geminiService.ts`. The service sends a prompt to Gemini with a system instruction that defines the output schema for scenes. Each scene includes a narrative, background style, duration, camera settings, and visual elements such as

3. Data model

`types.ts` defines the core structures: `VisualElement`, `Scene`, `AnimationScript`, and enums like `ElementType` and `BackgroundStyle`. Examples include `character`, `shape`, `text`, `map-marker`, `arrow`, `fade-in`, `slide-in`, `scale-up`, and `camera zoom/pull`.

4. Rendering and animation

`components/AnimationPlayer.tsx` renders scenes in a video-like viewport and handles progress timing, time-lapse, and scene transitions. Background styles include `default`, `map`, `grid`, `space`, and `paper`, and elements can `move`, `fly in`, `float`, and `pull out`. The player also displays subtitles, camera motion, and scene badges to mimic an educational video preview.

5. Project structure

`App.tsx`: UI and flow orchestration

`components/AnimationPlayer.tsx`: playback renderer

`services/geminiService.ts`: Gemini API integration and JSON schema enforcement

`types.ts`: data contracts

`vite.config.ts`: dev server and environment variable setup

6. Running locally

Requirements: Node.js, a Gemini API key, and npm dependencies.

Install: `npm install`

Set `GEMINI_API_KEY` in `.env.local`

Start: `npm run dev`; Vite serves the app on port 3000 by default

7. Overall assessment

This project is a compact, design-first AI app that turns educational prompts into visually structured, animated content. It is best described as a frontend-driven AI storyboard and animation generator for educational content creation.